High Accumulation and Subcellular Distribution of Thallium in Green Cabbage (Brassica Oleracea L. Var. Capitata L.).

The accumulation of thallium (Tl) in brassicaceous crops is widely known, but both the uptake extents of Tl by the individual cultivars of green cabbage and the distribution of Tl in the tissues of green cabbage are not well understood. Five commonly available cultivars of green cabbage grown in the Tl-spiked pot-culture trials were studied for the uptake extent and subcellular distribution of Tl. The results showed that all the trial cultivars mainly concentrated Tl in the leaves (101~192 mg/kg, DW) rather than in the roots or stems, with no significant differences among cultivars ($p = 0.455$). Tl accumulation in the leaves revealed obvious subcellular fractionation: cell cytosol and vacuole >> cell wall > cell organelles. The majority (~ 88%) of leaf-Tl was found to be in the fraction of cytosol and vacuole, which also served as the major storage site for other major elements such as Ca and Mg. This specific subcellular fractionation of Tl appeared to enable green cabbage to avoid Tl damage to its vital organelles and to help green cabbage tolerate and detoxify Tl. This study demonstrated that all the five green cabbage cultivars show a good application potential in the phytoremediation of Tl-contaminated soils.